13.4.7 Properties of isocyanides

Physical properties

- They are colourless, highly unpleasant smelling volatile liquids and are much more poisonous than the cyanides.
- They are only slightly soluble in water but are soluble in organic solvents.
- They are relatively less polar than alkyl cyanides. Thus, their melting point and boiling point are lower than cyanides.

13.4.8 Chemical properties

1) **Hydrolysis**: Alkyl isocyanides are not hydrolysed by alkalies. However they are hydrolysed with dilute mineral acids to give primary amines and formic acids.

$$CH_3 - NC + 2H_2O \xrightarrow{\text{acid}} CH_3 - NH_2 + HCOOH$$

Methyl isocyanide — Methylamine — Formicacid

2. **Reduction:** When reduced catalytically (or) by nascent hydrogen, they give secondary amines.

$$CH_3 - NC + 4[H] \xrightarrow[\text{(or) } Ni / H_2]{Na / C_2H_5OH} CH_3 - NH - CH_3$$

Methyl isocyanide — Dimethylamine

3. **Isomerisation:** When Alkyl isocyanides and heated at 250^0C, they change into the more stable, isomeric cyanides

$$CH_3 - N \equiv C \xrightarrow[250^0C]{\text{Heat}} CH_3 - C \equiv N:$$

Methyl isocyanide — Methylcyanide

4. **Addition reaction**. Alkyl isocyanides add on halogen, sulphur, and oxygen to form the corresponding addition compounds.

a) $R - N \equiv C + X_2 \longrightarrow R - \ddot{N} = C\begin{matrix} X \\ X \end{matrix}$

b) $R - N \equiv C + S \longrightarrow R - \ddot{N} = C = S$

Alkyl isothiocyanate

c) $R - N \equiv C + O \xrightarrow[\text{(or) } O_3]{2HgO} R - \ddot{N} = C = O + Hg_2O$

Alkyl isocyanate

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